

A Novel Detector Geometry to Measure the Muon Anomalous Magnetic Moment

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Abstract

The measurement of the muon anomalous magnetic moment is a precise test of the validity of the Standard Model of particle physics. Previous measurement of the anomaly shows a significant discrepancy between theory and experiment, signaling possibilities of new particles and subatomic forces. To verify and build upon those measurements, new proposed experiments must take more data to measure the anomaly to a higher precision. By increasing the rate of muons in the experiments and thus the amount of data collected, however, a new problem arises. Because of the presence of too many particles to be individually detected, the systematic error caused by pileup inhibits a precise measurement of the anomaly. A different method of taking data that does not deal with individual measurement is therefore proposed to solve the problem of pileup, but this method requires a detector geometry that optimizes its figure of merit (FOM). The optimization of the detector geometry FOM also minimizes the experimental systematic and statistical uncertainties, resulting in less run time and a more precise measurement. In this research, a Monte Carlo simulation has been developed to calculate the FOM of the detector geometry based on the number, asymmetry, and momentum of the positrons decayed from muons in the experiment. The simulation has been validated by comparing its calculation of the average asymmetry with the experimental value. We have designed and simulated a novel detector geometry with one calorimeter and shielding that optimizes the FOM. This detector geometry is optimized to accept particles in the experiment that contain the most data about the anomaly, and does so to the highest precision among all other geometries tested, including previous and proposed experiment geometries.

1. Introduction

The muon anomalous magnetic moment, or the $g - 2$ of the muon, is the difference between the precession of the spin and the precession of the momentum of the muon in a storage ring. The recently measured value of $g - 2$ at Brookhaven National Laboratory (BNL) with 0.54 parts per million precision [1] deviates from the Standard Model prediction by a significant amount (greater than 3 sigma) suggesting the existence of unknown subatomic particles and forces, such as supersymmetry.

A greater precision measurement of the muon anomalous magnetic moment would signal with increased certainty any new physics beyond the Standard Model or constrain present theories of the nature of particle physics, depending on its agreement with Standard Model predictions. Newly proposed experiments at the Japan Proton Accelerator Research Complex (JPARC) [2] and other laboratories [3, 4] to measure $g - 2$ offer a independent measurement that tests the Standard Model with unprecedented precision (0.1 parts per million) and the accuracy of previous experiments [2]. To address the challenges faced by these new experiments, a new detector geometry must be designed and optimized for a new method of taking data.

Recent muon $g - 2$ experiments at BNL, as shown in figure 1, have used storage rings to store muons at a "magic" momentum, 3.1 GeV, where the muon spin procession is unaffected by the electric focusing field [1]. Stored muons originate from a proton beam extracted from a synchrotron: the proton beam hits a nickel target and produces pions, which decay into the muons injected into the storage vacuum ring. The muons are stored in an ultra-uniform magnetic field and focused with an electrostatic field so that they circulate around the ring. As the polarized muons circulate in the ring, the spin precession of the muon starts to differ from the momentum direction (from when it was polarized along the momentum direction). Because the

muon is unstable, after a few microseconds in its laboratory frame, it decays into two neutrinos and a positron: the latter's decay direction, time, and momentum is a direct measurement of the anomalous magnetic moment [1, 2].

When a positron is emitted, it curls towards the center of the ring and hits a detector (calorimeter: at BNL, lead-scintillator type), which records information about 100 (BNL) to 10,000 (JPARC) positrons per storage ring fill [1, 2]. Due to the high number and rate of positrons being detected, a systematic error, pileup, is introduced. Pileup, or the almost simultaneous ($< 5\text{ns}$ apart) arrival of positrons at a detector, is the cause of incorrect measurements of positron momentum [1]; for example, two 0.8 GeV positrons arriving at the same time will be measured as one 1.6 GeV positron. Pileup accounts for $\sim 40\%$ of the total systematic error for the BNL experiment [1,5]. (Note: because a positron's mass is very small, its momentum is very similar to its energy).

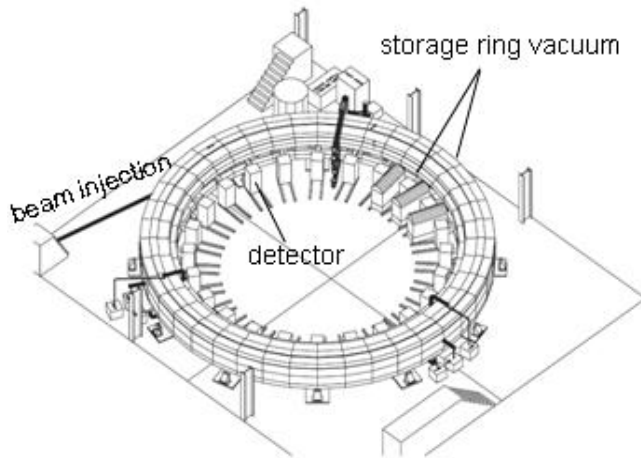


Figure 1. 3D representation of the BNL experiment. The wide outer ring is the storage ring. Muons injected at 10 O'clock in the representation decay into a positron and two neutrinos, which are detected on the inside of the storage ring by 24 equally spaced detectors, represented by tall, thin boxes.

Any new $g - 2$ experiment that seeks to improve upon previous ones will do so by one or more of the following: 1. store more muons, 2. introduce a higher rate of muons, 3. use an ultra-cold muon beam or 4. increase the number of storage ring fills per second. By using an ultra-cold muon beam, as JPARC proposes to do, measuring $g - 2$ becomes much simpler, and the storage

ring radius can be minimized to 33 cm; however, more muons must be stored and injected at a significantly higher rate [2].

These higher numbers and rates lead to higher pileup errors. Since the experiment at JPARC will need 500 times the muon rate of previous experiments [2], and other new experiments will possibly need the same high rate, pileup will become a major source of error if old methods, namely, the detector geometries and pileup correcting algorithms, are applied.

To address the issue of pileup, the 'Q method' has been developed [6, 7] in which the total sum of positron momentum over the run time is measured, as opposed to previous methods, where *individual* positron information was collected and recorded over time. Pileup ceases to be a problem with the Q method because a number of positrons, e.g. two 0.8 GeV positrons, arriving at the same time are added to the *total momentum sum* of positrons over the total time, without measuring individual positron numbers and momentum. However, the Q method requires a unique detector geometry with a high figure of merit because the statistical uncertainty of the measurement of the anomalous precession frequency is inversely proportional to the figure of merit of the detector geometry [1]: a greater figure of merit means less data collection and cost of experiment. The figure of merit of a detector geometry for the Q method is NA^2x [7], where N is the *number*, A is the *asymmetry*, and x is the *momentum*, of the positrons detected.

A Monte Carlo simulation would be the best means of testing this geometry, and a unique one is required to fit the parameters of the experiment in question. Detectors, positrons, the storage ring, and figure of merit are all represented in a way applicable to previous muon $g - 2$ experiments but flexible enough so that they could be applied to future experiments. After the simulation had been developed and verified, a number of detector geometries were designed and

tested. The focus of this paper will be on the development of the simulation and the search for an optimal detector geometry.

2. Methods

We have developed a comprehensive and accurate Monte Carlo simulation with detailed parameters relevant to the optimization of detector systems. The simulation is written in C++, and the results are visualized with a data plotting program, gnuplot. The simulation is developed uniquely to find the optimal detector geometry according to the following algorithm:

1. Design a detector geometry.
2. Generate positrons that fit experimental parameters using Monte Carlo methods.
3. For each positron generated, check for a hit with the detector geometry.
4. After all positrons are evaluated, print the figure of merit of the detector geometry.
5. Repeat 1- 4 with a new detector geometry.

The validity of the program was verified in a number of ways: comparing program results with experimental results, checking program results with graphical visualizations, and trigonometric tests.

2.1 Overall Parameters

The simulation is written to be a two-dimensional representation of the BNL experiment but general enough to fit any muon $g - 2$ experiment, e.g. Experiment B. Using basic coordinate geometry, trigonometry, and Monte Carlo methods (using many random events to determine the probability of a certain outcome), the simulation calculates the figure of merit for different detector geometries.

2.2 Detectors

A basic coordinate system is set with the origin at the center of the storage ring; the radius of the storage ring is set to 1. A detector is represented in this coordinate system using its vertices. For simplicity, the program calculates all points in the detector with the spacing between the points being 0.01, and stores detectors as a set of coordinates points, much like a computer represents a letter as a set of pixels, as shown in figure 2.

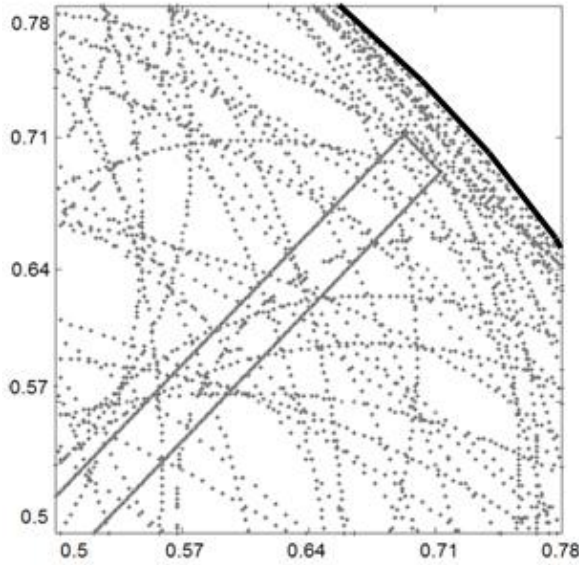


Figure 2. Schematic of detector and positrons *looking down on* a section of the storage ring (the unit is the storage ring radius). The straight, sloped lines represent the sides of an example detector through which positrons are detected; the curved sets of points are positron coordinates. To the top right is the storage ring boundary, shown in bold. Over a million coordinate pairs are calculated for the sides of the detectors and positrons paths.

2.3 Positrons

When a muon decays inside the storage ring of a $g - 2$ experiment, the resulting positron curls towards the center of the ring in a circular path. Each positron has a unique radius, point of decay, and asymmetry. The simulation generates 30,000 random positrons paths with defined radii, center coordinates, and angle of rotation. In figure 3, 100 such positrons are shown. The radii of the positrons represent their fractional momentum, e.g. the small circles are positrons with low momentum while the high momentum positrons have large radii; a positron with radius 0.4 has a momentum fraction of 0.4.

In the parametric form, the path of a positron with radius R , center x-coordinate h , and center y-coordinate k is:

$$Px = R \times \cos(t) + h \quad (1)$$

$$Py = R \times \sin(t) + k \quad (2)$$

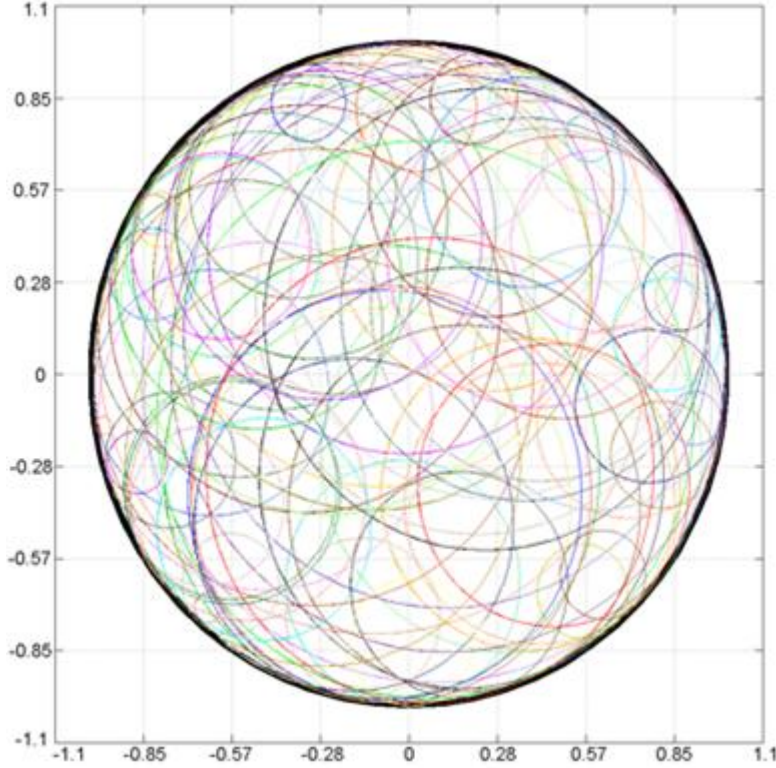


Figure 3. 2D *top* view of the muon storage ring with 100 positron paths. Each individual circle is a positron's decay path; in the simulation, each positron is assigned a unique and random decay point (where the circle is tangent to the storage ring), radius, and center coordinate pair. Higher momentum positrons have larger radii, and lower momentum positrons have smaller radii. The storage ring radius = 1.

where radius R can be set to the fractional momentum x , generated based on the distribution of the laboratory frame number N (see figure 4). Center coordinates (h,k) can be calculated based on the point of decay. A positron decayed from the point $(0, 1)$ has center coordinates $h = 0$ and

$$k = 1 - R \quad (3)$$

For a positron decayed from any point along the storage ring, we calculate h and k as:

$$h = -(1 - R) \sin \theta \quad (4)$$

$$k = (1 - R) \cos \theta \quad (5)$$

where θ is the rotation angle. Since positron paths can be tangent to any point along the storage ring circumference, θ is generated from a random number between 0 and 2π with uniform distribution.

The fractional positron momentum, x , is [8, 9]:

$$x = \frac{P}{P_\mu} \quad (6)$$

Here P is the momentum of the positron, and P_μ is the muon momentum, or maximum momentum. The normalized laboratory frame number $N(x)$ as a function of fractional positron momentum is derived in [8] and [9] as:

$$N(x) = \frac{(1-x)(5+5x-4x^2)}{5} \quad (7)$$

The program generates positrons to meet distribution in (7) (many positrons with low momentum, few with high momentum) by finding the roots of the normalized integral of $N(x)$:

$$I(x) = \frac{5}{3} \int N(x) = \frac{x^4 - 3x^3 + 5x}{3} = U \quad (8)$$

where U is a uniform random number in the range of [0,1]. Here we use the root of the function $I(x)$ as the normalized radius (R) of the positron. As seen in figure 4, the program generated laboratory frame number distribution is very close to the curve of $N(x)$ (error < 1%).

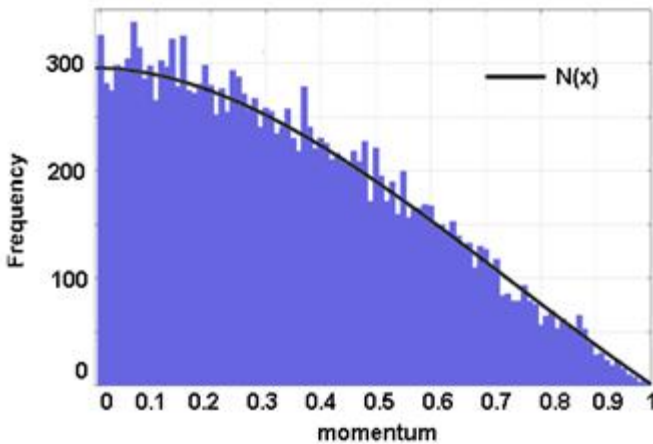


Figure 4. Comparison between the $N(x)$ curve and the histogram of program generated $N(x)$ using Monte Carlo integration. The distribution of program generated $N(x)$ for 30,000 positrons (binwidth = 0.01) is within 1% of the derived distribution of $N(x)$.

Each positron is thus generated with a specific radius, center, and angle of rotation. About 30,000 positrons are stored in an array in the same manner as the detectors. For each positron, after its radius, center, and angle of rotation is calculated according to the above mentioned methods, its general equation is represented in coordinate points as seen in figure 2.

2.4 Detection

As seen in figure 2, neither positron paths nor detector sides are solid, but rather they are a set of points. For each positron, the simulation checks to see if the positron path intersects with any detector sides. If both the x and y values for the detector and positron are within an error value, (in the simulation error = 0.01), a hit is recorded: in figure 2, if a coordinate set of both the detector and a positron are within 0.01 coordinate, there is a hit. After a hit is detected, the positron is discarded and a new one is tested, thus no positron is detected by more than one detector (the calorimeter absorbs the positron energy).

3. Results

An optimal detector geometry is one that optimizes the figure of merit (FOM):

$$FOM = NA(x)^2x \quad (9)$$

where $A(x)$ is the asymmetry [8, 9]:

$$A(x) = \frac{8x^2 - x - 1}{5 + 5x - 4x^2} \quad (10)$$

As stated in [1], the precision of measuring the anomalous precession frequency is proportional to the FOM. With a higher FOM a shorter data taking period is required to precisely measure the anomalous precession frequency.

Figure 5 shows $N(x)$, $A(x)$ and $FOM(x)$. It is seen that the highest FOM occurs in the range of fractional momentum 0.65 - 0.95, although only a simulation calculated 14.7% of all positrons fall within that range. Furthermore, all positrons with fractional momentum below 0.6 have low asymmetries and result in a low figure of merit. The optimal detector geometry should therefore minimize acceptance of positrons with fractional momentum less than 0.6 and maximize acceptance of positrons with fractional momentum greater than 0.65.

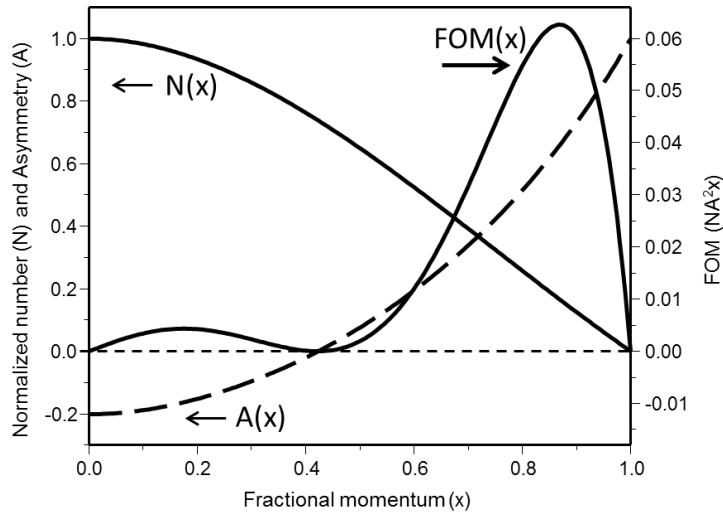


Figure 5. Normalized lab frame number $N(x)$, asymmetry $A(x)$ (represented as a dashed line), and $FOM(x)$ as a function of fractional positron momentum, x [1]. The horizontal dotted line marks the zero value. $FOM(x)$ depends heavily on $A(x)$ and is optimized at a high positron momentum. Notice at $x = 0.42$, $A(x) = 0$ and $FOM(x) = 0$. The optimal detector has high geometrical acceptance of positrons in the range $0.65 < x < 0.95$ (high FOM) and low geometrical acceptance of positrons in the range $x < 0.6$ (low FOM).

3.1 Program Verification

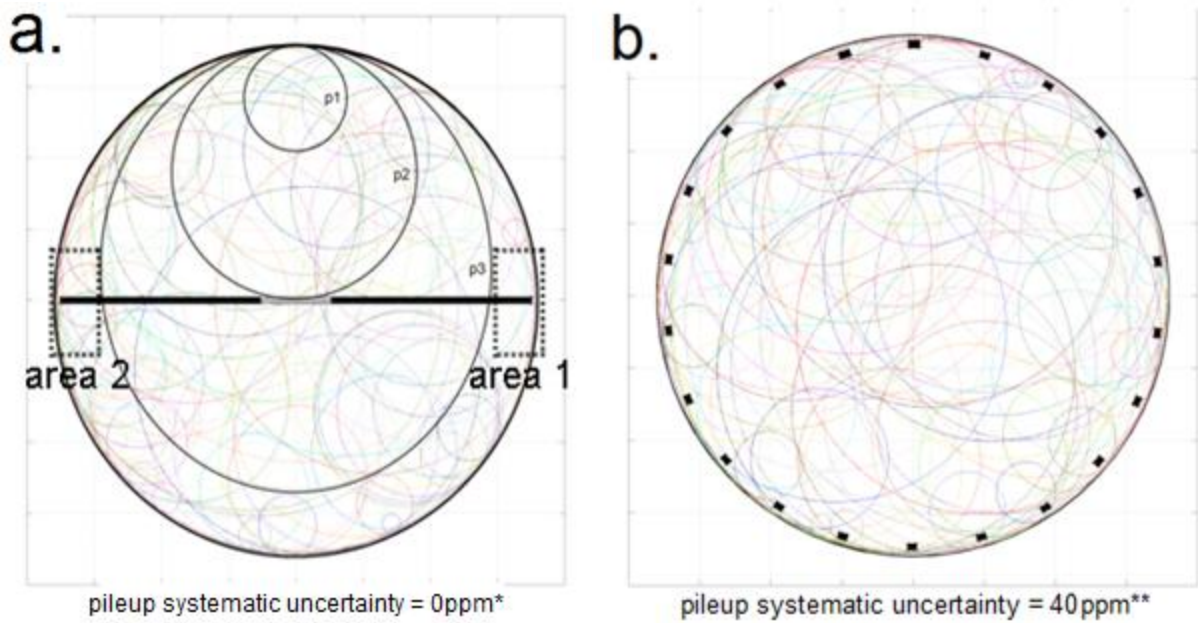
We check the validity of the program in a number of ways. The positrons and detector geometries were piped to a graphing program and checked visually and positrons that intersected with the detector were counted and compared to the program data for 50, 100, and 150 total positrons. The detector geometry was rotated by 90, 180, and 270 degrees, and the change in the figure of merit was found to be within 2% for each rotational trial. Most convincingly, when we simulate the detector geometry and data acquisition method of the BNL experiment, we calculate the asymmetry to be 0.39, which is very close to the experimental value of ~ 0.4 [1, 10].

3.2 Detector Optimization

Our first detector geometry design (table 1: #1) is a simplistic long and thin detector positioned at the center of the storage ring with the size of $1.98r_0 \times 0.02r_0$ (r_0 being the radius of the storage ring), as shown in figure 6. This design would minimize acceptance of most positrons with low momentum, e.g. the positron 'p1' in figure 6 will not hit the detector due to the thin detector design. The minimum thickness of the detector is $\sim 0.02r_0$ so as to completely measure the positron momentum. The detector accepts most positrons with medium to high momentum, including those in the range of 0.3 - 0.6.

The positrons with medium momentum, however, have very low asymmetry, and thus result in low FOM. To minimize the acceptance of these positrons, we introduce a gap breaking the single detector into two, which would reduce detector acceptance in the momentum range 0.3 - 0.6, e.g. 'p2', as shown in figure 6. The detector size is thus $0.82r_0 \times 0.02r_0$ for each of the two detectors (table 1: #2). It is noted that the right and left ends of the detector (area 1 and 2) accept not only the low momentum positrons ($x < 0.2$) decayed near the detector but also high momentum positrons ($x > 0.65$), such as 'p3,' and are thus necessary.

A more carefully examination, however, reveals that almost all high momentum positrons such as 'p3' ($x > 0.65$) have decay paths that hit both detectors, as shown in figure 6. Low momentum positrons, however, hit only one detector, mostly in areas 1 and 2 in figure 6. One detector, e.g. the left detector, we thus remove with little effect on the acceptance of high momentum positrons, as shown in Figure 7a. Its dimensions are optimized as $0.77r_0 \times 0.02r_0$ (table 1: #3). A detector geometry design with one detector is therefore better than a design with two detectors, not only because the cost is reduced, but also because acceptance of low momentum positrons is reduced in half (area 2 is eliminated), further increasing the FOM.



*Using Q method eliminates pileup systematic uncertainty.

**Using the data analysis method in [1] and other previous $g - 2$ experiments with a proposed 1.6GHz muon rate [2] leads to 500 times greater pileup systematic uncertainty ($uncertainty = rate \times timing\ resolution$) if the timing resolution remains constant.

Figure 6. a) Top view of the storage ring with detectors and positrons. Three theoretical positrons are shown: p1 with a small radius, p2 with a radius fraction 0.5, and p3 with a large radius. Two detectors are shown in solid, separated by a gray gap, optimized in length to minimize acceptance of positrons with momentum fraction 0.3 - 0.6, e.g. p2. b) BNL geometry as shown in figure 1, program generated. Twenty-four detectors are shown in solid, equally spaced around the storage ring. If such a detector geometry is used with proposed experiment rates at JPARC [2] the resulting pileup systematic uncertainty would equal 40ppm (parts per million) or 4,000% of the acceptable systematic error for a 0.1ppm measurement of $g - 2$.

In another effort to minimize surface area for acceptance of low momentum positrons, the detector was sloped inward at the tips and undesirable positrons near the edge and center of the storage ring were reduced (figure 7b). However, this geometry is flawed because at the tips, the calorimeter lacks the depth required for the calorimeter to measure a positron's momentum.

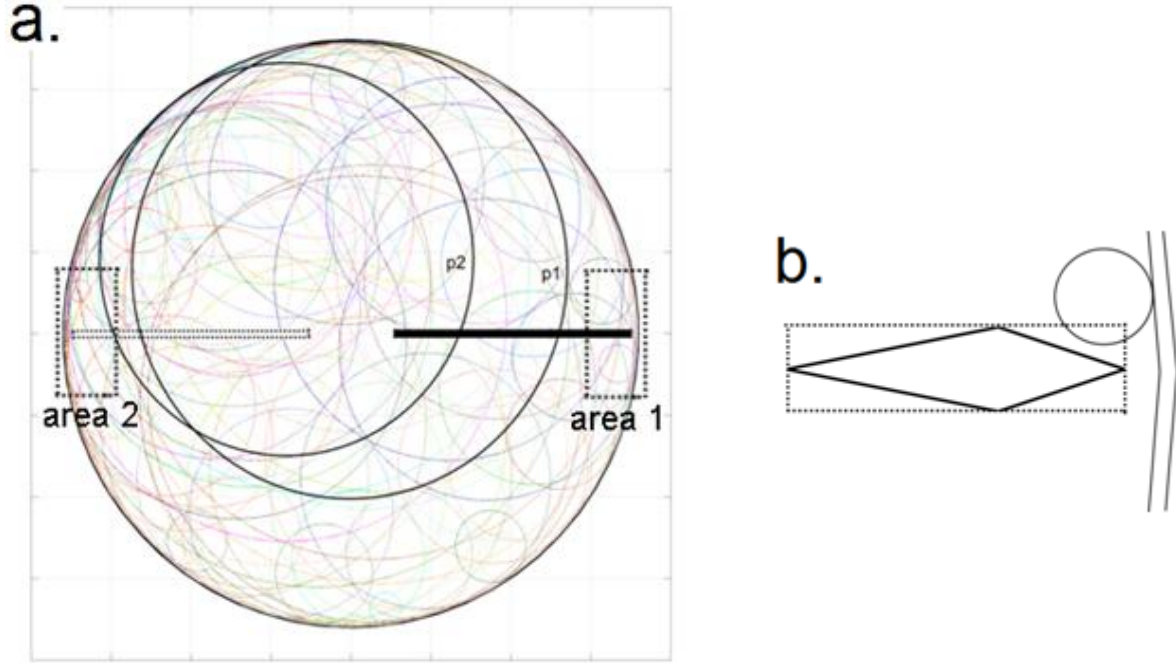


Figure 7. a) Improved detector geometry: one detector (one detector shown in solid, the other with dotted lines) will accept almost all high momentum positrons that two would have because the positron decay paths are circular. The area for low momentum positron acceptance is therefore halved (area 2 is eliminated).

b) Magnified view of a detector with sloped ends: this geometry minimizes acceptance of low momentum positrons but at the tips lacks sufficient depth to contain the energy of a high momentum positron entering the calorimeter.

3.3 Shielding Optimization

To further improve the figure of merit, we add a layer of shielding on one side of the detector. The shielding is optimized to absorb all positrons $x < 0.6$, thus eliminating them from detection. Requirements for optimizing the shielding include 1. the shielding must have sufficient "stopping power," $\frac{dE}{dx}$, to absorb most low momentum positrons (for JPARC, the highest energy being 300MeV [2]), 2. the material used must have a low magnetic susceptibility as to not interfere with the ultra-uniform magnetic field, and 3. the shielding must not affect high momentum positron acceptance.

Low momentum positrons lose momentum when travelling through matter mainly as a function of $\frac{dE}{dx}$. High momentum positrons lose momentum mainly by *bremsstrahlung*: emitting radiation as a function of the radiation length of the shielding material.

At first, we considered the shielding to be most useful if placed around the inside of the storage ring, thus absorbing positrons as soon as they left the ring. However this setup was not practical because the thickness of the shielding was limited in the design (at JPARC, the calorimeter is 0.46cm away from the storage ring [2]). It would not have sufficient stopping power, e.g. lead shielding with a thickness of 0.4cm only stops positrons with energy less than 9 MeV. Moreover, because of *bremsstrahlung*, high momentum positrons leaving the ring would scatter in an unfavorable direction, perhaps leaving the storage ring.

As seen in figure 8, because muons always travel clockwise in the storage ring, a simulation calculated ~90% of the low momentum positrons detected hit only one side (down side, in the figure: side A) of the calorimeter, and ~98% of the high momentum positrons detected hit the other side (up side: side B). Shielding might be placed on side B while not interfering with the high momentum positrons. If the distance between the detector and shielding is small and the thickness of the shielding is restricted to two radiation lengths, the radiation given off as a positron travels through the shielding will travel towards the detector. We limit the thickness to two radiations lengths to minimize the probability of the *bremsstrahlung* radiation missing the detector. When this limitation is accounted for, however, even the optimal shielding material, copper, with a radiation length of 1.5cm [11] and a stopping power of 11 MeV/ cm [11] will only stop positrons with $x < 0.1$, which just slightly *decreases* the FOM of the detector geometry. Therefore we discard shielding on side B.

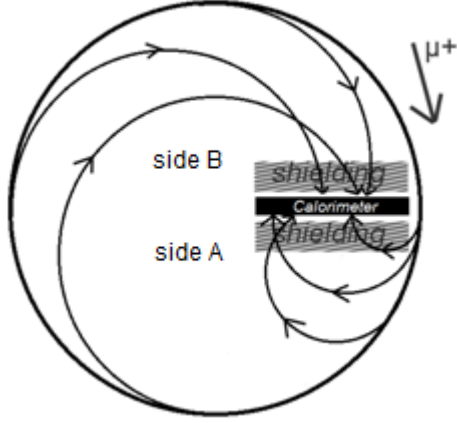


Figure 8. Two pieces of shielding on either side of the detector. Since the muons always travel clockwise, low momentum positrons primarily hit the calorimeter on one side, while high momentum positrons hit the other side.

We derive the thickness of shielding, $T(x)$, needed to absorb all positrons below x as:

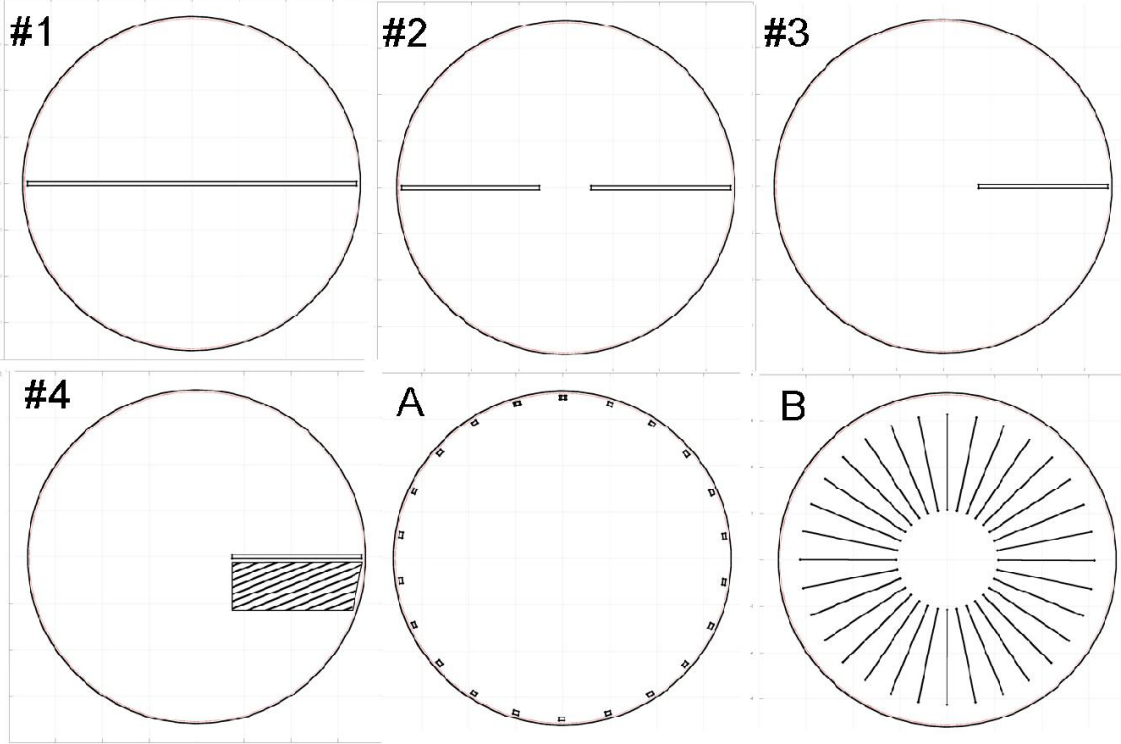
$$T(x) = \frac{x}{S \times \pi P} \quad (11)$$

where x is the familiar momentum fraction, πP is distance travelled by a positron through shielding (half the circumference) where P is the positron momentum, and S is the stopping power:

$$S = \frac{dE}{dx} \div P_\mu \quad (12)$$

Since *low* momentum positrons lose little energy due to *bremsstrahlung*, radiation length can be ignored when choosing a shielding material with high $\frac{dE}{dx}$ on side A. We choose tungsten, which has a $\frac{dE}{dx}$ of 21.1MeV/cm and a magnetic susceptibility of $7.1 \times 10^{-10} m^3 mol^{-1}$ [11], as the optimal material for this shielding. To block all positrons with momentum fraction $x < 0.6$, by (11), the tungsten shielding must be $0.16r_0$ thick and lie parallel to side A of the detector, as shown in figure 8 and 9. This addition to the detector geometry (table 1: #4) increases the figure of merit by 281% compared to the same detector geometry without shielding.

Table 1. A series of designed detector geometries are shown below. The corresponding N , $\bar{A}$, $\bar{x}$, and FOM for the Q method are listed in the table. The simulation calculated N is the fractional number of positrons detected, $\bar{A}$ is the average asymmetry of positrons detected, and $\bar{x}$ is the average momentum. The calculated FOM is thus the average FOM for all positrons detected. The optimal detector geometry designed in this paper has the highest FOM among any previously used detector geometry and 197% greater FOM than the JPARC proposed detector geometry.



Detector	N	$\bar{A}$	$\bar{x}$	FOM for Q Method
1	0.451	0.111	0.530	0.00285
2	0.434	0.115	0.539	0.00309
3	0.281	0.158	0.571	0.00401
4	0.142	0.339	0.687	0.0112
A (geometry used in [1])	0.551	0.0159	0.439	0.0000619 ^a
B (geometry proposed in [2])*	0.818	0.0832	0.316	0.00566 ^b

^a This FOM is significantly lower for the Q method because its detector geometry was optimized for a different method of analysis and different experimental conditions.

*This geometry is simulated accurately as described in [2]: both the simulation and experimental detectors measure positron momentum. The simulated calorimeter absorbs all positrons $0.3 < x < 0.6$ after half their paths are measured.

^b This FOM is calculated as NA^2 [1, 2] because its detector geometry does not use the Q method to deal with pileup. The FOM for the Q method of this geometry is 0.00179.

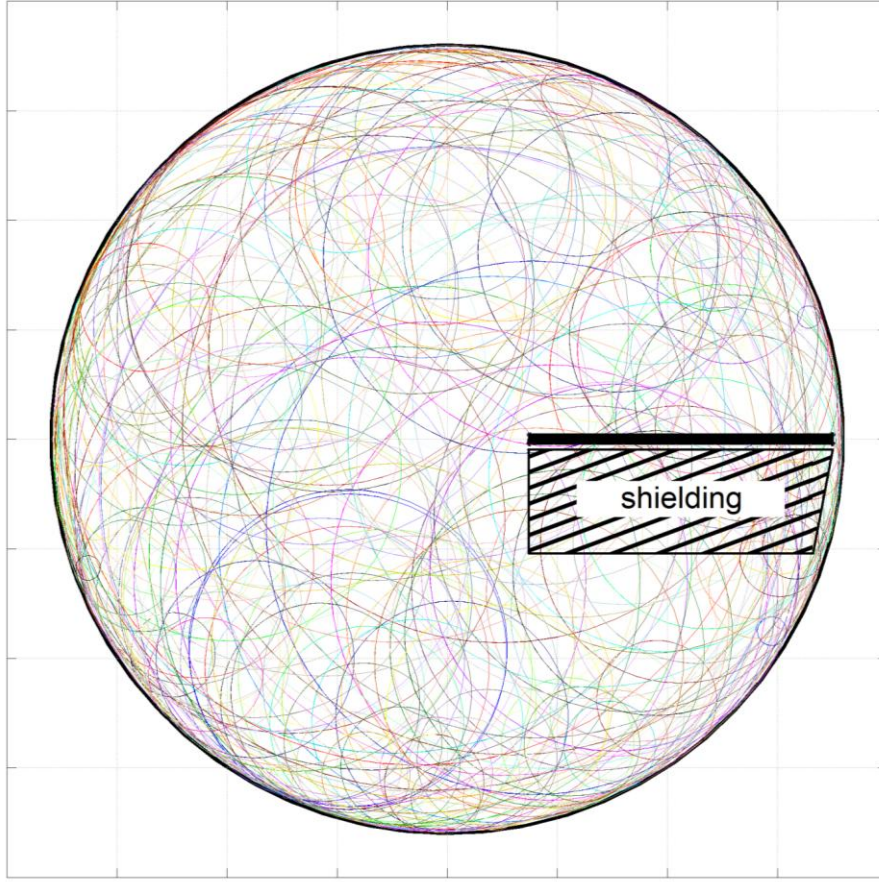


Figure 9. The optimal detector geometry. The shielding and detector are represented in striped and solid, respectively. A shielding material placed parallel to the length of the detector will eliminate a portion of low momentum positrons from being detected.

4. Discussion

Any $g - 2$ experiment that seeks to make a greater precision measurement of $g - 2$ will have increased muon rates which result in an increased systematic uncertainty due to pileup if old detector geometries and data taking methods are applied [2]. Previous detector geometries optimized for lower muon rates will experience pileup at extremely high levels (400 times the proposed JPARC precision measurement) at JPARC or any new $g - 2$ experiment. The detector geometry designed here (figure 9), however, is optimized for the Q method and is not sensitive to pileup. The optimization of the detector geometry for the Q method by increasing the FOM reduces the statistical error due to the detector geometry, thus reducing the run time and cost of a $g - 2$ experiment. In addition, because the detector geometry dimensions designed here are given

in terms of the radius of the storage ring, the detector geometry can apply to any $g - 2$ experiment. This is possible because both the positron and maximum momentum can be expressed in terms of the storage ring radius.

The geometry presented here would be simpler and more practical than proposed detector geometries to deal with high muon rates, especially the JPARC proposed detector geometry. JPARC proposes to measure $g - 2$ with 64 silicon strip tracking detectors placed radially around the inside of the storage ring to detect positron paths and derive momentum [2]. A calorimeter is placed at the center to absorb positrons with momentum 0.3 - 0.6, lowering their contribution to the data volume (figure 10) [2].

However, the proposed silicon tracker geometry remains to be tested for its feasibility. It is unclear whether crucial aspects of its design to contain pileup uncertainty, such as a high segmentation or fast readout ($< 1\text{ns}$ [2]), will work. Furthermore, the proposed design has lower calculated FOM than the detector geometry we have designed.

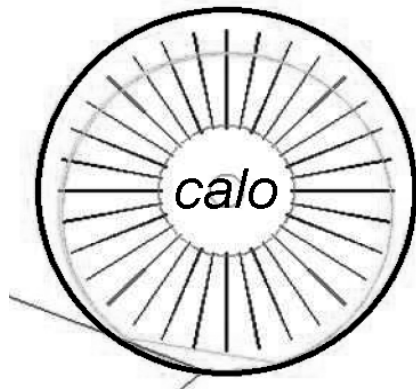


Figure 10. JPARC proposed detector geometry. Top view of 64 silicon strip detectors which detect the path of positrons as they pass through the tracking strips. Crucial aspects of this geometry, such as high segmentation and faster than 1ns readout, have yet to be tested for reliability or feasibility. At the center is a calorimeter to absorb positrons with momentum fraction $0.3 < x < 0.6$.

The optimal geometry designed here addresses the weaknesses that the proposed geometry fails to address. It does not rely on untested solutions such as high segmentation or fast readout, but simply on the sum of all energies over a run period measured by a calorimeter. The detector we use in this geometry is a lead-scintillator calorimeter, which has been used in [1] and

[12], while the proposed silicon tracking detector in [2] has never been tested in a $g - 2$ experiment. Finally, our detector geometry has the highest FOM for the Q method among both the geometry used at BNL and the geometry proposed by JPARC.

5. Conclusions and Future Work

Future muon $g - 2$ experiments will face challenges in precisely measuring the muon anomalous magnetic moment. One of the best ways to address pile systematic error is to use the Q method of measuring data. An accurate simulation has been developed to calculate the figure of merit of different detector geometries. The accuracy of the simulation has been compared to experimental results and its validity confirmed.

We have designed a novel detector geometry that includes extensive shielding of the detector against low energy positrons. The detector to be used in this geometry is a long and thin calorimeter that is $0.77r_0 \times 0.02r_0$. Tungsten shielding placed on the side of the detector that accepts most of the low momentum positrons is optimized to be $0.16r_0$ thick. This geometry has a FOM 197% greater than the proposed JPARC geometry and 18,255% greater than the geometry used at BNL in proposed experimental conditions.

In the future, further work must be done to build the detector and data acquisition system in addition to testing the setup with a low intensity muon beam. We would be very interested to see the adoption of this geometry in future experiments and its realization as a crucial component in measuring the muon anomalous magnetic moment.

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